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A DSGR device records the electric current that results from applying a voltage source to the skin and underlying blood vessels. DSGR can be modeled by the circuit shown in Figure 1, in which V represents the applied potential, C_B represents the capacitance of the blood, and R_E , R_D , and R_B represent the electrical resistances of the epidermis, the dermis, and the blood, respectively.

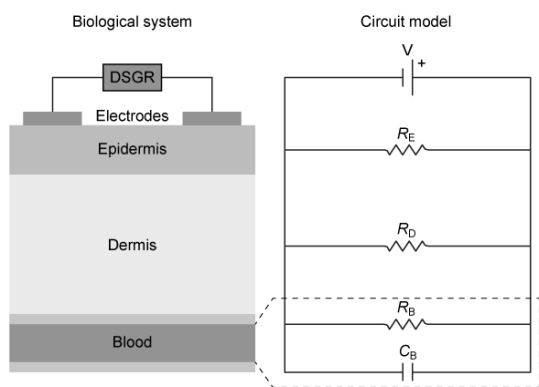


Figure 1 DSGR circuit model including skin and superficial blood vessel

The capacitance of charged blood may be explained by red blood cell membranes acting as physical barriers that separate electrons introduced into the blood from positively charged ions in the red blood cell cytoplasm. The dielectric constant k of red blood cell membranes varies with glucose concentration (Figure 2) because glucose uptake by red blood cells alters the activity of membrane-bound proteins that regulate the flow of ions into and out of the cell.

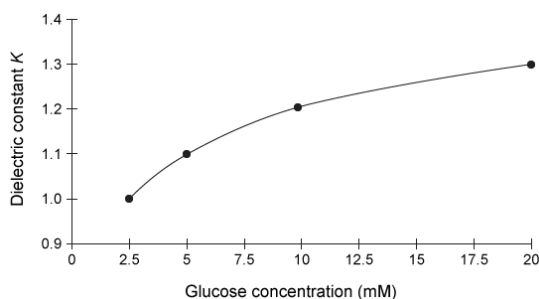


Figure 2 Blood dielectric constant vs blood glucose concentration

DSGR readings vary with blood glucose concentration because the time needed to fully charge the blood is related to total blood capacitance. However, interpreting DSGR readings may be complicated by changes in biological variables other than blood capacitance. For example, the resistivity of blood varies in accordance with osmolarity such that changes in diet or hydration status influence the current measured by DSGR devices.

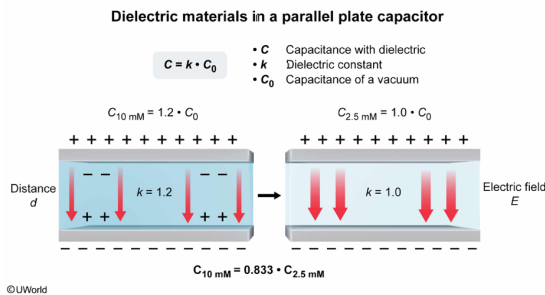
Suppose that a 5-mL blood sample with a glucose concentration of 10 mM has capacitance C . How will C change if the concentration of blood glucose is reduced to 2.5 mM?

- ✓ ☒ A. C will decrease by approximately 20%. [80%]
- ☐ B. C will remain the same. [5%]
- ☐ C. C will increase by approximately 20%. [10%]
- ☐ D. C will increase by approximately 40%. [3%]

Correct

81% answered correctly

Explanation:



Capacitors store electrical charge between equally and oppositely charged conductive surfaces. The simplest type of **capacitor** is the **parallel plate capacitor**, a device featuring conductive plate surfaces with a defined area (A) separated by distance (d). The strength of a capacitor is quantified by its capacitance (C), which is the ratio of the quantity of stored charge (Q) to the **voltage** (V) generated by the separated charges:

$$C = \frac{Q}{V}$$

The capacitance of a parallel plate capacitor is influenced by the material that separates the charged plates. Inserting **dielectrics** (polarizable materials) between the plates of a parallel plate capacitor causes electric dipoles to form within the dielectric material. These dipoles reduce the magnitude of the **electric field** (E) present between the plates such that voltage decreases and capacitance increases:

$$\uparrow C = \frac{Q}{\downarrow V} = \frac{Q}{\downarrow E \cdot d}$$

The **dielectric constant** (κ) reflects the ability of a dielectric to increase the capacitance of a parallel plate capacitor relative to the capacitance of a vacuum (C_0) (ie, nothing between capacitor plates):

$$C = \kappa \cdot C_0$$

When a DSGR device charges a sample of blood, the cellular membranes of red blood cells act as dielectric substances that separate equal and opposite charges on each side of the membrane. **Figure 2** shows that the dielectric constant of red blood cell membranes is 1.2 at a

blood glucose concentration of 10 mM and is 1.0 (equal to that of a vacuum) at a blood glucose concentration of 2.5 mM. Accordingly, the capacitance values for blood at these blood glucose concentrations are:

$$C_{10 \text{ mM}} = 1.2 \cdot C_0$$

$$C_{2.5 \text{ mM}} = 1.0 \cdot C_0$$

The change in capacitance between a blood glucose concentration of 10 mM and 2.5 mM can be compared by calculating the percent difference relative to the initial concentration:

$$\% \text{ difference} = \frac{(C_{2.5 \text{ mM}} - C_{10 \text{ mM}})}{C_{10 \text{ mM}}} \times 100$$

$$\% \text{ difference} = \frac{(1.0 \cdot C_0 - 1.2 \cdot C_0)}{1.2 \cdot C_0} \times 100$$

$$\% \text{ difference} = \frac{-0.2 \cdot C_0}{1.2 \cdot C_0} \times 100 = -17 \%$$

Therefore, decreasing the glucose concentration of blood from 10 mM to 2.5 mM will decrease the capacitance of red blood cell membranes blood by approximately 20% (**Choice A**).

Educational objective:

Dielectrics are polarizable substances that can be inserted between the plates of a parallel plate capacitor to increase its capacitance. The dielectric constant quantifies the ability of a dielectric material to increase capacitance.

Time spent: 0 sec

Subject:
Physics

QID: 400348

System:
Electrostatics and Circuits

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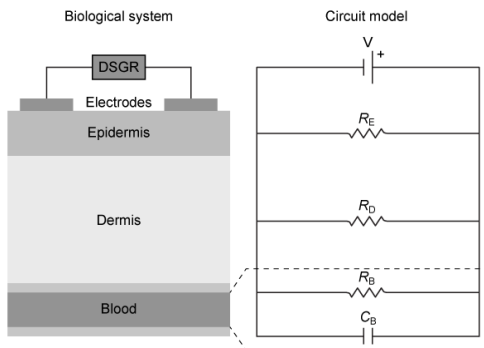


Figure 1 DSGR circuit model including skin and superficial blood vessel

The capacitance of charged blood may be explained by red blood cell membranes acting as physical barriers that separate electrons introduced into the blood from positively charged ions in the red blood cell cytoplasm. The dielectric constant k of red blood cell membranes varies with glucose concentration (Figure 2) because glucose uptake by red blood cells alters the activity of membrane-bound proteins that regulate the flow of ions into and out of the cell.

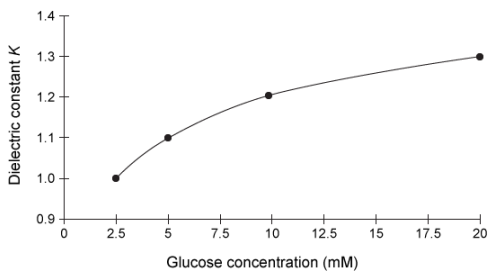
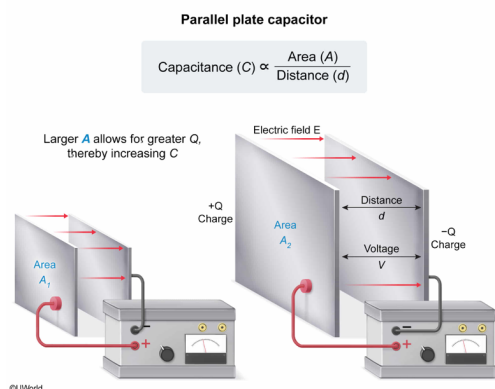


Figure 2 Blood dielectric constant vs blood glucose concentration

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Assuming that the size and shape of red blood cells are uniform, by what factor would the quantity of red blood cells need to change to raise the capacitance of a blood sample to 250% of its original value?

- ☐ A. $2/5$ [6%]
- ☐ B. $(2/5)^2$ [8%]
- ✓ ☒ C. $5/2$ [60%]
- ☐ D. $(5/2)^2$ [23%]

Explanation:

A **parallel plate capacitor** stores electrical charge (Q) on two equally but oppositely charged regions separated by some distance (d). **Capacitance (C)** is equal to the ratio of Q and the voltage (V) across the capacitor. Furthermore, because V equals the product of **electric field strength E** and d , it follows that C is **inversely proportional** to d :

$$C = \frac{Q}{V} = \frac{Q}{E \cdot d} \Rightarrow C \propto \frac{1}{d}$$

Capacitance also depends on the **physical dimensions** of each plate, because a higher Q can be stored without generating **repulsive electric forces** with a larger plate area (A). Hence, C is also **directly proportional** to A :

$$C \propto A$$

Therefore, the overall proportionality expression relating capacitance to area and distance is:

$$C \propto \frac{A}{d}$$

In the context of **DSGR**, red blood cells act as a physical barrier that separates negatively charged electrons in the blood plasma from positively charged ions in the red blood cell cytoplasm. As a result, the cumulative surface area of the red blood cells in a blood sample is directly proportional to the total capacitance of the blood sample.

Because the surface area of each red blood cell is generally considered to be fixed and uniform, increasing the capacitance of a blood sample requires increasing the quantity of red blood cells. Therefore, increasing the capacitance of blood to 250% (ie, 2.5 times its original value or by a factor of $5/2$) would require increasing the quantity of red blood cells by a factor of $5/2$.

(Choice A) The capacitance of the blood is proportional to the total surface area of the red blood cells. Therefore, changing the quantity of red blood cells by a factor of $2/5$ will decrease the capacitance to 40% of the original value, not increase the capacitance to 250% of the original value.

(Choices B and D) The total surface area of the red blood cells is proportional to the number of red blood cells, not the square of the number of blood cells.

Educational objective:

The capacitance of a parallel plate capacitor is related to capacitor plate material, capacitor plate area, and the distance between plates. Capacitance is directly proportional to capacitor plate area and inversely proportional to the distance between plates.

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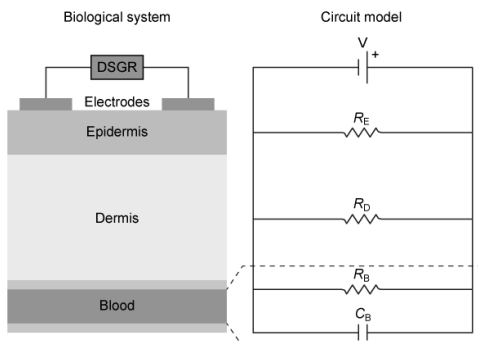


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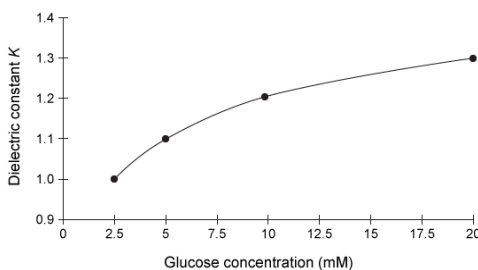


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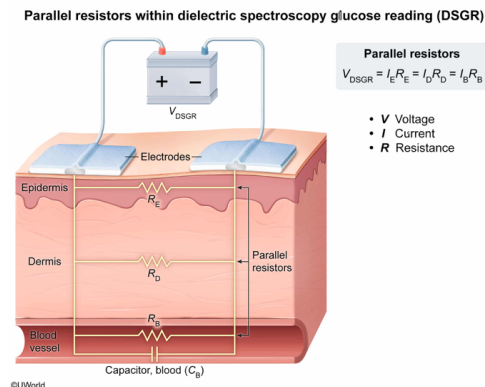
Which of the following statements best describes the behavior of the circuit in Figure 1 when the fully charged capacitor (comprising red blood cells) is discharged?

- ☐ A. The electric currents through R_B and R_E are equal because the resistors are in series. [3%]
- ✓ ☒ B. The voltage drops across R_B and R_E are equal because the resistors are in parallel. [76%]
- ☐ C. The electric current through R_B exceeds that through R_E because the capacitor is fully charged. [9%]
- ☐ D. Both the current and voltage drop across R_B and R_E are equal because the same power source is used. [10%]

Correct

76% answered correctly

Explanation:



The arrangement of components in **electric circuits** affects the overall behavior of the circuit and the electric phenomena that occur at each circuit component. For example, **series circuits** position electrical components in succession and form a single channel through which a fixed quantity of current passes. In contrast, **parallel circuits** position circuit components side by side and have multiple channels through which current can flow. These differences alter the behavior of circuit components.

For **resistors in series**, the current (I) passing through each resistor is constant because charge entering a resistor series cannot accumulate. As a result, the **voltage drop** associated with any given resistor results from its own intrinsic resistance R . However, the voltage drop that occurs over the course of a series of resistors is cumulative and equal to the source voltage (V):

$$V = IR_1 + IR_2 + \dots + IR_n$$

Resistors in parallel act as multiple channels through which current can flow. Consequently, the current traveling through each resistor varies in accordance with its own unique resistance. However, the voltage drop that occurs as current crosses each resistor is identical because all resistors experience the same electric potential from the source voltage:

$$V = I_1 R_1 = I_2 R_2 = I_n R_n$$

During DSGR, the biological resistances formed by the blood (R_B) and epidermis (R_E) oppose the flow of current between terminals of the DSGR device. R_B and R_E are arranged in parallel, so the voltage drop

across each will be identical to the voltage of the device (V_{DSGR}):

$$V_{DSGR} = I_B R_B = I_E R_E$$

(Choices A and D) For the current traveling across R_B and R_E to be identical, both resistors would either have to be placed in series or the resistance of each resistor would have to be identical. R_B and R_E are arranged in *parallel*, and the individual resistance of both resistors is unknown.

(Choice C) The current traveling through the blood is impacted by the value of R_B in relation to the other resistances within the circuit. No conclusions about the current can be reached without knowing the resistance values of all elements.

Educational objective:

Circuit components may be placed in series (positioned successively) or in parallel (positioned side by side). For resistors in parallel, the voltage drop across each resistor is identical. For resistors in series, the current passing through each resistor is identical.

Time spent:0 sec

Subject:

Physics

QID:400350

System:

Electrostatics and Circuits

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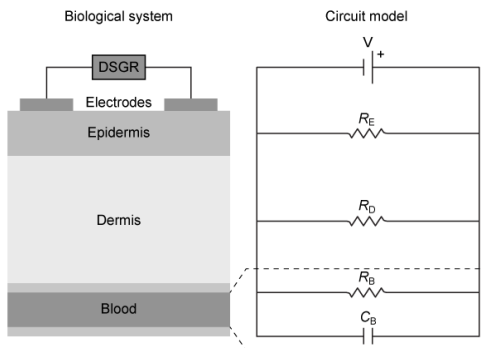


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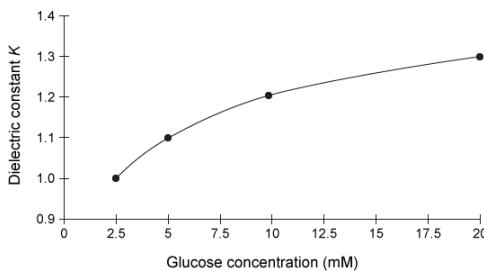


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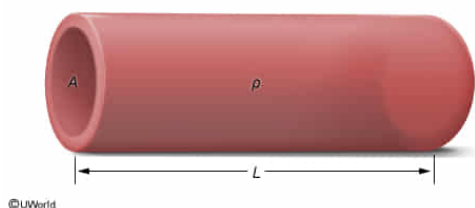
Assume that the average radial artery has a cross-sectional area A of 0.2 cm^2 and a length L of 30 cm . What is the total resistance R of the radial artery? (Note: The resistivity of blood is $200 \text{ } \Omega\text{-cm}$.)

- ☐ A. $3 \times 10^1 \text{ } \Omega$ [4%]
- ☐ B. $3 \times 10^2 \text{ } \Omega$ [15%]
- ☐ C. $3 \times 10^3 \text{ } \Omega$ [11%]
- ☒ D. $3 \times 10^4 \text{ } \Omega$ [68%]

Explanation:**Resistance of a resistor**

$$R = \rho \cdot \frac{L}{A}$$

- R Resistance
- ρ Resistivity
- L Length
- A Cross-sectional area



Ohm's law states that when electric current passes through an object, some electric potential energy (ie, voltage) will be lost due to some electric resistance within the object. As such, **resistance (R)** refers to the capacity of an object to oppose the movement of charge (ie, current) through the object. Resistance depends in part on the physical dimensions of the resistor. For example, resistance is **directly proportional** to the length (L) but **inversely proportional** to the cross-sectional area (A):

$$R \propto \frac{L}{A}$$

However, the resistance of two objects with the same physical dimensions can still differ if the resistors are made from different materials with distinct physical properties. Therefore, quantifying the relative differences in the resistance of different materials can be achieved by measuring their **resistivity (ρ)** (ie, the inherent resistance per unit length for a uniform cross section of the material), an intensive property that is directly proportional to resistance. Consequently, the overall resistance of an object can be expressed as:

$$R = \rho \cdot \frac{L}{A}$$

In this scenario, the cylindrical column of blood within the radial artery acts as a resistor to an electric current traveling from one end of the artery to the other. The question states that the average cross-sectional area and length of the radial artery are 0.2 cm^2 and 30 cm , respectively. Furthermore, the typical ionic content of blood has a resistivity of $200 \text{ } \Omega \cdot \text{cm}$. Therefore, the total resistance of the blood within the radial artery can be calculated using the equation above:

$$\begin{aligned} R &= (200 \text{ } \Omega \cdot \text{cm}) \cdot \frac{30 \text{ cm}}{0.2 \text{ cm}^2} \\ R &= (200 \text{ } \Omega \cdot \text{cm}) \cdot (150 \text{ cm}^{-1}) \\ R &= 30,000 \text{ } \Omega = 3 \times 10^4 \text{ } \Omega \end{aligned}$$

Therefore, the resistance of the column of blood within the radial artery is equal to $3 \times 10^4 \text{ } \Omega$ (**Choice D**).

Educational objective:

Resistance describes an object's opposition to the flow of electric charge (ie, current). The resistance of an object is directly proportional to its resistivity and length but inversely proportional to its cross-sectional area.

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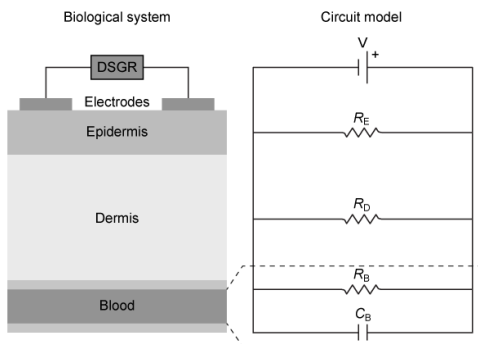


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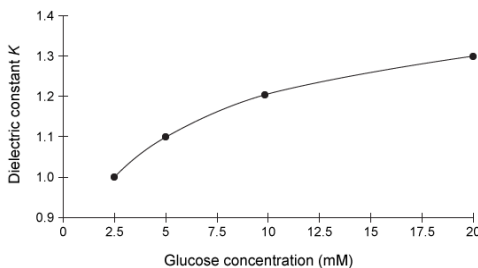


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To simplify calculations, mathematical models of DSGR can be generated that replace R_E and R_D in

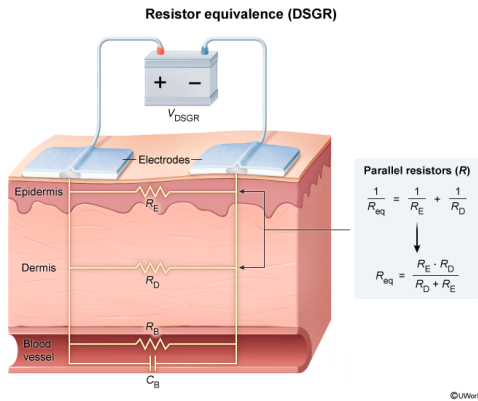
Figure 1 with an equivalent resistor R_{eq} . The mathematical expression for R_{eq} is:

- ☐ A. $R_E + R_D$ [14%]
- ☐ B. $R_E \cdot R_D$ [1%]
- ☐ C. $(R_E + R_D)/(R_E \cdot R_D)$ [31%]
- ☒ D. $(R_E \cdot R_D)/(R_E + R_D)$ [52%]

Correct

51% answered correctly

Explanation:



Resistors are electrical components that oppose the movement of charge (ie, current) and **deplete electric potential energy** (ie, voltage) according to **Ohm's law**. The ability of a resistor to oppose the movement of current is quantified by its **resistance (R)**, which is an extensive property (measured in ohms Ω) that is related in part to the physical dimensions of the resistor itself.

Like other circuit components, groupings of two or more resistors may be placed either in **series** or in **parallel** configurations. Arranging multiple resistors in series is the same as creating one equivalent resistor (R_{eq}) with a resistance equal to the sum of the resistances of each individual resistor:

$$R_{eq} = R_1 + R_2 + \dots + R_n$$

Conversely, placing two or more resistors in parallel yields an equivalent resistor (R_{eq}) with a resistance that is *less* than the resistance of any one resistor in the arrangement. Specifically, the value of the equivalent resistor is given by the following expression:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

In this scenario, two of the resistors within the DSGR circuit (R_E and R_D) are to be replaced with an equivalent resistor when modeling DSGR. R_E and R_D are parallel because they are located side by side and share the same voltage source. Therefore, the expression for R_{eq} is:

$$\frac{1}{R_{eq}} = \frac{1}{R_E} + \frac{1}{R_D}$$

Multiplying each term on the right side of the expression by a ratio equivalent to 1 gives each the same denominator:

$$\frac{1}{R_{eq}} = \frac{1}{R_E} \cdot \frac{R_D}{R_D} + \frac{1}{R_D} \cdot \frac{R_E}{R_E}$$

$$\frac{1}{R_{eq}} = \frac{R_D}{R_E \cdot R_D} + \frac{R_E}{R_E \cdot R_D}$$

$$\frac{1}{R_{eq}} = \frac{R_D + R_E}{R_E \cdot R_D}$$

Inverting both sides of the expression above yields:

$$R_{eq} = \frac{R_E \cdot R_D}{R_D + R_E}$$

Therefore, a resistor equivalent to R_E and R_D would have a resistance calculated as the product of R_E and R_D divided by the sum of R_E and R_D (**Choice D**).

Educational objective:

Resistors are circuit components that deplete electric potential energy by opposing electric current.

Placing multiple resistors in series increases the total resistance whereas placing multiple resistors in parallel decreases the total resistance.

Time spent:0 sec

Subject:
Physics

QID:400352

System:
Electrostatics and Circuits

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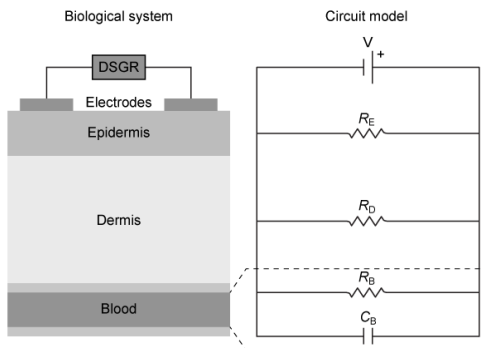


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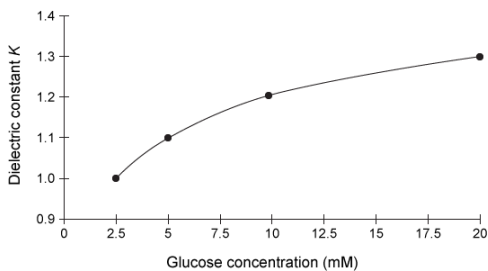


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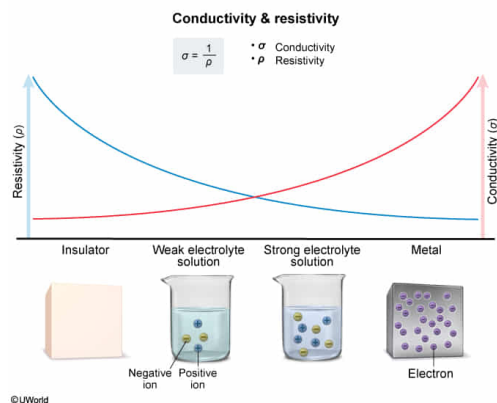
When dietary sodium intake increases, the electric current measured by a DSGR device also increases. Current increases because:

- ☐ A. blood pressure energy is converted into electric potential energy. [4%]
- ☐ B. the accumulation of positively charged ions causes blood glucose to precipitate. [10%]
- ☒ C. electrolytes within blood transport electric charge. [80%]
- ☐ D. alterations in red blood cell shape increase capacitance. [4%]

Correct

80% answered correctly

Explanation:



Conductivity (σ) is an intensive property that describes the ease with which charges flow through the molecular structure of a given material. Conductivity is directly proportional to the conductance of an **electric conductor** just as **resistivity (ρ)** is directly proportional to the resistance (R) of a resistor. However, conductivity is **inversely proportional** to resistivity; as such, a material described as having high conductivity will be characterized by low resistivity:

$$\sigma \propto \frac{1}{\rho}$$

Most metals are **highly conductive** whereas most nonmetals are nonconductive or only partially conductive (ie, semiconductors). Metals are highly conductive because valence electrons within metals are loosely associated with each metal nuclei such that the application of an electric field to a metal easily **dislodges electrons** and generates a current.

In contrast to solid objects, **electrolytic solutions** may be conductive in the absence of freely moving electrons. Instead, the conductivity of electrolytic solutions is most closely related to the **concentration of ions** because the motion of these charge carriers throughout the fluid generates current.

Consequently, increased dietary sodium intake contributes to an increased electrolyte (charge carrier) content within the blood. As a result, the resistivity of blood decreases while the conductivity of blood increases. Therefore, the electric current detected by **DSGR** increases with greater dietary sodium intake.

(Choice A) **Bernoulli's equation** states that blood pressure may be converted into **kinetic energy** or **gravitational potential energy** as a fluid flows between two points within a conduit. However, blood pressure cannot be readily converted into electric potential energy (ie, voltage).

(Choice B) Glucose is a neutral (uncharged) molecule at physiologic pH and does not contribute to the conductivity of blood. Furthermore, neutral molecules within a solution do not precipitate following the addition of charged ions.

(Choice D) Increased blood capacitance following the addition of sodium would divert the electric potential energy toward charging red blood cell membranes, *decreasing* the electric current passing between DSGR terminals through the blood.

Educational objective:

Conductivity describes the ease with which electric current flows through a material. The conductivity of metals may be attributed to the loosely associated valence electrons within a metal whereas the conductivity of electrolytic solutions is directly proportional to the molar concentration of charged ions.